Co-Designing Support for Colour-Coded Technical Data with a Colour-Blind Engineering Student

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Abstract: Engineering students rely on colour to read graphs and simulation results, which is a problem for the roughly 8% of men with red-green colour vision deficiency (CVD). We worked with one co-designer, Guilherme, a 5th-year Mechanical Engineering MSc student with mild red-green CVD, across three co-design sessions. In the first session we interviewed him and ran a questionnaire. In the second session we confirmed our problem statement with him, brainstormed ideas, and sorted idea cards into want, maybe and no. In the third session he tried existing tools and a Wizard of Oz prototype of a handheld colour lens. A tooltip showing the colour under a crosshair helped him most, and he found the lens easier than asking a colleague. Based on his suggestion, our final design is a handheld bar with a lens that pops out at the press of a button. We contribute an account of how one colour-blind student reads technical data, and a co-designed concept that works without installing any software.

CCS Concepts: • **Human-centered computing → Accessibility technologies**; *Participatory design*; Empirical studies in accessibility.

Additional Key Words and Phrases: colour vision deficiency, co-design, inclusive design, data visualisation, engineering education

1 Introduction

Colour is used everywhere in engineering to show data: simulation results, graphs, maps and software interfaces. For people with red-green CVD, many of these colours look the same. About 8% of men and 0.5% of women have a red-green deficiency [Colour Blind Awareness], so a typical class of 30 students has at least one [Institute of Physics 2020].

The problem is made worse by common design choices. Rainbow colour maps and red-green pairs are still widely used, and they can distort the data for everyone, not only for people with CVD [Crameri et al. 2020]. Students with CVD report that they spend time working out what a figure shows instead of learning from it [Institute of Physics 2020].

Existing aids do not fully solve this. Most of them recolour the screen or name a colour, which still leaves the user to match that colour to the legend [Geddes et al. 2023; Fiorentini]. Software tools also work only on the computer where they are installed.

We decided to work with one person instead of a general group. Guilherme is a Mechanical Engineering student who uses SolidWorks and PrusaSlicer in his thesis. After the first session we narrowed our problem to this statement, which he signed at the start of session 2 (see projectStatement-signed.pdf):

"Colour-blind engineering students like Guilherme need to match the colours in technical graphs and simulation colour scales to the values they represent, in order to interpret their own results independently."

Over three co-design sessions, we explored this problem with him, generated ideas together, and tested existing tools and a prototype. The result is a handheld lens that he can point at any screen to see the colour under a crosshair. Its final form, a bar from which the lens pops out at the press of a button, was proposed by Guilherme himself.

This paper makes three contributions. First, it shows that for this student the problem is reading colour-coded data, not seeing colours. Second, it shows that a tooltip naming the colour at one point helped him more than a filter that recolours the whole screen. Third, it presents a co-designed concept that works on any screen, including lab computers where he cannot install software.

2 Related Work

CVD in Science and Engineering. CVD is common, and it is not rare in STEM classes. Guidance from the Institute of Physics says that colour-only graphs, diagrams and software are among the main barriers for these students [Institute of Physics 2020]. Crameri et al. [2020] also show that rainbow colour maps can distort data by more than 7% for all readers.

Existing Aids. Most existing tools fall into four groups: colour-correcting glasses and lenses [Almustanyir et al. 2026], software that recolours the screen [Microsoft DirectX Developer Blog 2023], apps that name colours [Fiorentini], and symbol systems such as ColorADD [International Colour Orchestration and Design]. Colour-correcting glasses showed no measurable improvement in standard colour vision tests with 48 participants [Gómez-Robledo et al. 2018]. Recolouring tools are often designed from simulations of CVD, not from the experience of people who have it [Geddes et al. 2023]. In session 3 we tested two of these tools with our co-designer (Section 3.4).

Co-Design. Co-design means that the people who will use a solution take part in designing it. We followed the Double Diamond model and worked with one co-designer through three sessions, so our results are about his needs and not about all people with CVD. Other HCI projects work in a similar way. Piedade et al. [2022] co-designed a wearable display with one person living with dissociative identity disorder, using storyboards, card sorting and

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paper prototypes. Tong et al. [2024] ran participatory design with practitioners over six months to set the requirements for a VR stage lighting course.

Gap. We found many tools that work on a screen or on a photo, but little about students reading colour scales in technical software. This is the gap our work addresses.

3 Methodology

We followed the Double Diamond design framework (*Discover*, *Define*, *Develop*, *Deliver*) across three co-design sessions over seven weeks. Session 1 explored the problem, session 2 explored possible solutions, and session 3 tested existing tools and a Wizard of Oz prototype of our concept with the co-designer.

3.1 Co-designer Profile

Our co-designer, Guilherme Quinteiro, is a 5th-year MSc student in Mechanical Engineering who is completing his master's thesis. He has mild red-green colour vision deficiency (CVD), likely deuteranomaly or protanomaly. We recruited him through personal contacts of the team and he took part voluntarily.

3.2 Exploring the Problem Space

The goal of session 1 was to understand how CVD affects Guilherme's everyday life and engineering work. We started by explaining the scope and goals of the project, so that he knew what we were asking and why.

We then ran a semi-structured interview that combined a 17-item questionnaire with open questions. The questionnaire measured where colour confusion happens in his life, how often it happens in his engineering work, how much time and effort it costs him, and how severe the consequences of a misreading can be.

The open questions let him describe his experiences in his own words, including his workarounds: inferring values from visual trends, guessing, or asking peers. We used these answers to write the problem statement in Section 1, and compared it with the literature on CVD in STEM education (Section 2).

The main implication for the next stage was that the problem is reading colour-coded data on screens, not seeing colours in the lab. Session 2 therefore focused on solutions for reading screen data.

3.3 Exploring the Solution Space

The goal of session 2 was to move into the second diamond (*Develop*) and explore solutions with the co-designer. We started with a brainstorm in which Guilherme and the team proposed ideas side by side, without judging them.

We then ran a card sort with cards for existing products, possible solutions, and blank cards for new ideas. Guilherme sorted the cards into *Want*, *Maybe* and *No*. His top three were a colour-naming app (like Color Blind Pal [Fiorentini]), a point-to-read tool that shows the value or colour name of a selected spot, and glasses or another type of wearable.

Session 2 also showed that he had never tried any assistive tool, although colour affects his work often. He explained that each case felt too small to look for a tool. This set the main requirement for the next stage: any solution must take less effort than asking a colleague. In this session, we photographed the brainstorm and the card-sorting.

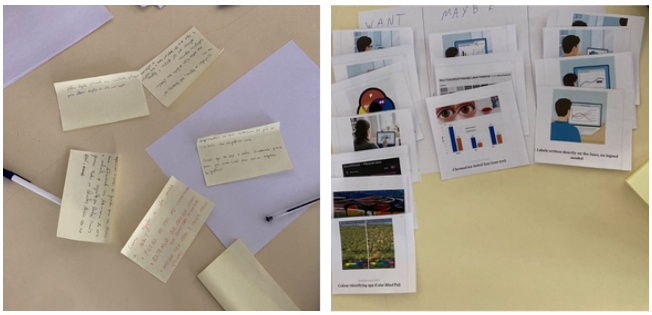


Fig. 1. Session 2: the brainstorm (left) and the card sort (right).

3.4 Prototyping and Feedback

The goal of session 3 was to test our concept against tools that already exist, in the *Deliver* phase. Before the session we filled in the first half of a prototyping report card from the IDEO Design Kit [IDEO.org] (see prototypeReportCard.pdf). It set three learning questions: whether the tooltip lets him read a plot without asking anyone, whether picking up a lens is easier than the existing tools or asking a colleague, and which overlay he needs (a tooltip at one point, a filter over the whole screen, or both).

To establish a baseline, the co-designer evaluated existing software tools, specifically Windows 11 deuteranopia colour filters [Microsoft DirectX Developer Blog 2023] and the Microsoft PowerToys Color Picker utility [Microsoft 2025] which shows the name of the colour next to the cursor while reading a SolidWorks stress simulation from his thesis workflow.

We then tested our concept, a handheld colour lens, as a Wizard of Oz prototype, so that we could test the experience without building the technology. To test the user experience prior to technical implementation, a research member manually simulated the device's internal display and real-time colour recognition simulating its display and its colour recognition. To explore the form, the team 3D printed several versions of the lens with different handle shapes and lens frame sizes, and he tried each one. The tooltip was shown to him on our personal laptops screen, while the 3Dprinted proto types were given to him in hand.

As in the previous sessions, we took written notes during the session, then read them together as a team and grouped the main points. We used these points to make the final changes to the prototype, which led to the final design (Section 5). In this session we photographed the 3D-printed prototypes (see Appendix A), and the **Wizard of Oz video:** `https://drive.google.com/drive/folders/1zkFqQ-hedk8ucPCQvwZc_qW_JVUeh3Yo`

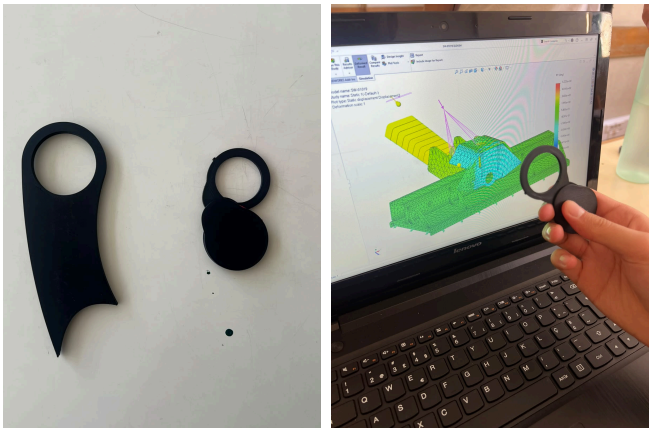


Fig. 2. Session 3: the prototypes (left) and the co-designer testing (right).

4 Findings and Design Criteria

The co-designer's challenge is screen-based rather than physical. We first assumed it concerned laboratory equipment, but his main barrier is reading colour-coded technical data: SolidWorks simulations, PrusaSlicer legends, and multi-line graphs. Similar hues (red/green, dark blue/purple, green/brown) are harder to tell apart under warm light. He copes by inferring values, guessing, or asking peers, but he rated these workarounds as helping only a little. Since asking a colleague takes under a minute and he does not mind it, any solution must be faster and simpler than asking for help.

Through ideation and card sorting, the co-designer prioritized three concepts (Section 3.3): a colour-naming app (like Color Blind Pal [Fiorentini]), a point-to-read tool, and wearable. He was also comfortable using a solution that others can see. These preferences pointed to a tool that shows the colour or value at a chosen point on a screen, with less effort than asking a colleague.

A software tool alone does not fully meet this need. Guilherme works on several computers, including lab computers where he cannot install software. We therefore kept the point-to-read idea but moved it into a physical object that he can carry and use on any screen. In session 2 we first imagined this object as a wearable, such as glasses. We chose a handheld lens instead because he already wears glasses: carrying a second pair and switching between them, or wearing one over the other, would be more constraining than helpful.

Session 3 answered the three learning questions on our report card. First, the tooltip let him read colours without asking anyone. He named it the most helpful feature, both in the PowerToys Color Picker and in our prototype. Second, he found using the lens easier than asking a colleague, which until now was his main way of coping. Third, the deuteranopia filter also helped him, but less than the tooltip, so the tooltip should be the main feature and the filter an optional aid. The metrics we used were qualitative, mainly his own feedback on this experience when compared to trying the same thing without the aid.

The existing tools helped him more than we expected. Since he had never tried any tool before, this suggests that the barrier was finding and setting up a tool, not the quality of the tools themselves.

On form, the co-designer did not settle on any of our printed shapes. Instead, he proposed his own: a handheld bar with the lens stored inside, which slides out at the press of a button, like a pop-out knife. He proposed this form factor because an exposed lens could easily be damaged in a lab environment. The pop-out mechanism protects the optics, improves portability, and accommodates potential internal electronics within a larger handle. Although the hardware remains speculative, we adopted his structural feedback directly into the final design (Section 5).

5 Final Design

The final design follows the co-designer's suggestion. It is a handheld bar with a lens that pops out at the press of a button and slides back in after use. The user aims the crosshair at any point on a screen or printed figure, and the lens shows a tooltip with the colour under it. An optional deuteranopia filter can be turned on through the lens.



Fig. 3. The final prototype design (left) and the 3D-printed version (right).

The design is a concept: the session tested the experience, not the technology. We propose that the bar holds a small camera behind the lens, a microcontroller that identifies the colour at the crosshair, a small display that shows the tooltip, and a rechargeable battery. In a future version, the device could also match the colour to the legend of the figure and show its value, which is what our problem statement asks for. Building this hardware is future work.

6 Discussion and Limitations

The PowerToys Color Picker already gave him a useful tooltip, so our design has to offer more than it does. The lens works away from his own computer: on lab computers where he cannot install software, on a colleague's laptop, on projected slides and on printed figures. It also needs no shortcut to remember, which matters for a user who never looked for a tool because each case felt too small to bother.

Our work has three main limitations. We worked with one co-designer, so the results describe his needs and not those of all people with CVD. The prototype was Wizard of Oz, so we did not test whether a camera can read screen colours accurately,

especially under the warm light he finds hardest. Finally, the final design has not yet been tested with him.

7 Conclusion

We co-designed a tool with Guilherme, a colour-blind engineering student, to help him read colour-coded technical data. Across three sessions we found that his problem is not seeing colours but matching them to the values in graphs and simulation legends, and that any solution must take less effort than asking a colleague. In session 3, a tooltip naming the colour at one point helped him more than a filter that recolours the whole screen. The final design, a handheld bar with a pop-out lens that he proposed himself, brings this tooltip to any screen, including lab computers where he cannot install software. The next steps are to build the hardware, add matching of colours to legend values, and test the device with him in his own work.

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A 3rd Co-Design Session

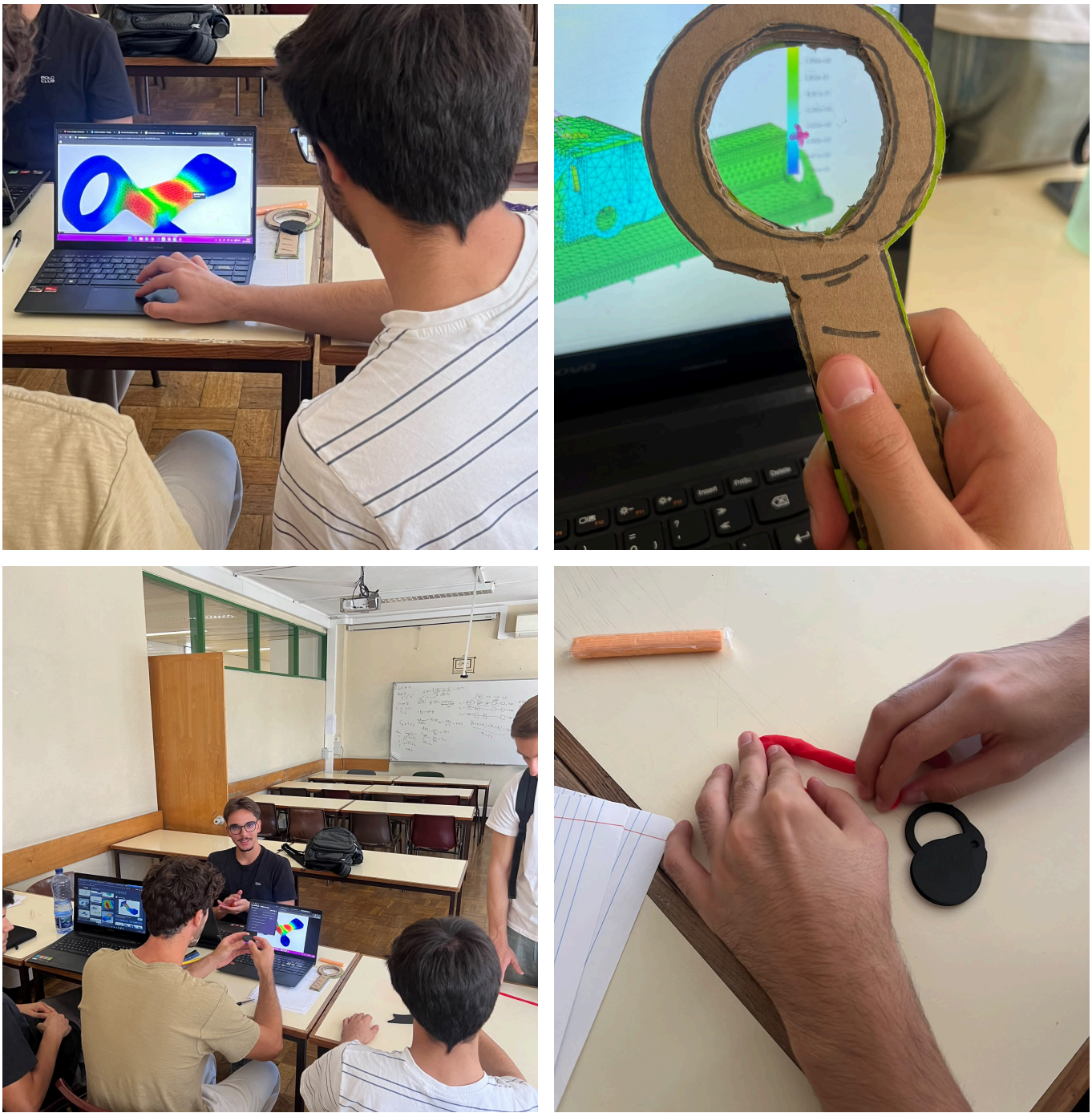


Fig. 4. The third co-design session: the co-designer (top left), a rough prototype (top right), the session (bottom left) and the play-dough model (bottom right). **Link to the 3rd session and final Wizard of Oz video:** https://drive.google.com/drive/folders/1zkFqQ-hedk8ucPCQvwZc_qW_JVUeh3Yo